From NDC Targets to Production Networks: Macroeconomic Spillovers of Brazil's Energy Transition

Summary

Achieving Brazil's Nationally Determined Contribution (NDC) under the Paris Agreement requires not only setting emissions targets, but also understanding how the reallocation of economic activity propagates through the production system. While integrated assessment models (IAMs) are central to informing climate targets, they offer limited insight into how transition-related demand shocks interact with intersectoral linkages and generate macroeconomic spillovers. This paper addresses this gap by linking IAM-informed mitigation pathways with a demand-driven input–output framework tailored to the Brazilian economy. Using outputs from the Brazilian Land Use and Energy System (BLUES) model, the analysis translates NDC-transition pathways into reallocations of final demand across existing sectors. These shocks are propagated through the production network using an input–output model that captures direct and indirect effects without imposing equilibrium assumptions. Baseline results highlight the systemic importance of energy-intensive sectors, particularly petroleum refining, oil and gas extraction, chemicals, and utilities, which exhibit high production multipliers (up to 2.4), large income multipliers (up to 7.4), and forward linkage extraction losses exceeding 4%, despite low direct labor intensity. Scenario results indicate that an NDC-aligned reallocation of demand increases total gross output by approximately 0.3% in 2030 and 1.2% in 2050 relative to baseline, alongside employment gains of around 180 thousand jobs in 2030 and 420 thousand jobs in 2050, and labor income gains exceeding 1% by mid-century. Overall, the findings show that the economic implications of Brazil's NDC are shaped by inherited production linkages, underscoring the need for climate policy to be complemented by industrial and labor-market strategies.

1. Introduction

Achieving national climate commitments requires more than defining long-term mitigation targets; it requires an understanding of how climate-aligned investments and structural change interact with existing economic systems. Over the past decade, integrated assessment models (IAMs) have become the most frequently used analytical tools for evaluating decarbonization pathways, offering projections of emissions trajectories, energy system transformations, and cumulative investment requirements (Riahi et al., 2017; Rogelj et al., 2018; IPCC, 2023). As a result, IAMs play a central role in informing Nationally Determined Contributions (NDCs) and shaping national and global climate strategies.

However, despite their strengths, IAMs are not designed to assess how mitigation pathways propagate through economies. By construction, they rely on highly aggregated representations of economic activity and production structures, which limit their ability to capture intersectoral linkages and macroeconomic dynamics (Grubb et al., 2015; Mercure et al., 2019). Consequently, while IAMs provide robust estimates of what needs to be deployed to meet climate goals, they offer more limited insight into how transition-related investment and demand shocks are transmitted across industries.

This limitation is particularly relevant from a policy perspective. Energy transitions involve large reallocations of final demand towards infrastructure, capital goods, low-carbon energy systems, and enabling services and away from carbon-intensive activities (Bataille et al., 2021; IEA, 2023). The macroeconomic effects of these reallocations depend not only on their aggregate magnitude, but also on the structure of the production network through which they propagate. Thus, differences in upstream and downstream linkages, labour intensity, and intermediate input requirements imply heterogeneous sectoral responses (Miller and Blair, 2009; Dietzenbacher et al., 2013).

In this context, Brazil offers a particularly relevant case. The country combines ambitious climate commitments with a distinctive emissions profile in which agriculture, forestry, and land use (AFOLU) account for roughly 70% of total greenhouse gas emissions, while fossil energy use remains structurally embedded in the production system. Brazil's updated NDC commits to a 53% reduction in greenhouse gas emissions by 2030 relative to 2005 levels and to achieving net-zero emissions by 2050 (Brazil, 2023). Although a substantial share of mitigation is expected to arise from land-use change and agricultural practices, meeting these targets also requires a progressive reduction in fossil-related activities and a reorientation of energy and industrial systems. At the same time, the Brazilian economy exhibits persistent intersectoral linkages to oil, gas, and bioenergy across multiple activities (IBGE, 2020; Alves-Passoni, 2020). Understanding how NDC-aligned transition pathways interact with this inherited production structure is therefore essential for anticipating macroeconomic and socio-economic outcomes.

Most existing macroeconomic assessments of climate policy in Brazil and elsewhere rely on equilibrium-based approaches such as computable general equilibrium (CGE) models (Böhringer et al., 2012; Lucena et al., 2016; Pereira et al., 2020). While these frameworks provide valuable insights into price adjustments and long-run equilibrium outcomes, they typically assume smooth reallocation mechanisms and full market substitution. Nonetheless, such assumptions may obscure the structural transmission channels through which transition-related shocks unfold (Storm, 2017; Mercure et al., 2018). From a policy perspective, this limits their usefulness for identifying where adjustment pressures are likely to concentrate and which sectors play a disproportionate role in amplifying or absorbing transition dynamics.

Against this background, this paper adopts a complementary perspective by focusing explicitly on the structural propagation of climate transition pathways through the production network. Rather than modelling equilibrium outcomes, the analysis addresses a policy-relevant question: given today's intersectoral linkages, how does an NDC-consistent energy transition translate into sectoral and aggregate economic effects? To address this question, the paper links IAM-informed mitigation pathways with a demand-driven input–output representation of the Brazilian economy.

Input–output models are particularly well-suited to tracing how exogenous demand shocks propagate through production networks via direct and indirect effects, without imposing equilibrium assumptions (Leontief, 1936; Miller and Blair, 2009). Recent studies have increasingly combined energy-system outputs with input–output frameworks to assess the macroeconomic and distributional implications of low-carbon transitions (Wiebe et al., 2019; Guan et al., 2020; Pollitt et al., 2021). Building on this

literature, the present analysis provides an assessment of Brazil's NDC-aligned transition in terms of macroeconomic indicators.

The remainder of the paper is organized as follows. Section 2 presents the methodological framework, detailing the integration of IAM-informed transition pathways with a demand-driven input–output model. Section 3 reports the results, and Section 4 concludes by discussing the implications of the findings for industrial strategy and the policy design.

2. Methodological Framework

This study employs a soft-link framework that integrates long-term transition insights from an Integrated Assessment Model (IAM) with a demand-driven input–output (IO) model to assess the macrostructural implications of alternative energy transition pathways. The IAM provides information on the magnitude, timing, and sectoral orientation of structural change required to meet Brazil's climate commitments, while the IO framework captures how such changes propagate through the existing production structure via interindustry linkages, without imposing equilibrium assumptions.

2.1. IAM framework and climate targets

IAM inputs are drawn from the Brazilian Land Use and Energy System (BLUES) model, a national integrated assessment model that represents interactions between the Brazilian energy system, land use, and greenhouse gas emissions under long-term technological and policy constraints (Rochedo et al., 2018; Lucena et al., 2016). BLUES has been used by Brazilian institutions to assess mitigation options and to support the formulation of Brazil's Nationally Determined Contribution, by evaluating the scale and system-wide implications of alternative emissions targets (Schaeffer et al., 2021).

Brazil's updated NDC commits the country to a 53% reduction in greenhouse gas emissions by 2030 relative to 2005 levels and to net-zero emissions by 2050 (Brazil, 2023). With total emissions of approximately 2.8 $GtCO_2e$ in 2005, these targets imply an emissions level of roughly 1.3 $GtCO_2e$ by 2030, followed by a rapid decline thereafter. Within this context, BLUES provides mitigation pathways that translate these policy targets into trajectories for energy system transformation, infrastructure expansion, and the deployment of carbon dioxide removal (CDR) options.

2.2. Input–output framework

The economy-wide propagation of transition-related changes is assessed using a demand-driven input–output model of the Brazilian economy. The analytical framework follows the structural IO approach developed by Alves-Passoni (2021). This framework is particularly suited to analyzing large-scale structural

reallocations, as it represents intersectoral dependencies and indirect effects along supply chains.

Formally, sectoral gross output in the year t is determined by the standard Leontief quantity model:

$$x_t = (I - A)^{-1} y_t$$

where x_t is the vector of sectoral gross output, A is the matrix of technical coefficients derived from the Brazilian input–output table, y_t is the vector of final demand, and $(I-A)^{-1}$ is the Leontief inverse. Scenario effects enter the model through changes in final demand, Δy_t, while technical coefficients are held constant.

Baseline structural characteristics of the economy are examined using standard IO indicators, including production multipliers, backward and forward linkages, fields of influence, and hypothetical extraction analysis (Rasmussen, 1956; Hirschman, 1958; Dietzenbacher and Los, 1998; Dietzenbacher and van der Linden, 1997). These diagnostics characterize the structural conditions under which transition-related demand shocks propagate, providing a benchmark for interpreting scenario results.

2.3. Translating IAM-informed pathways into IO demand shocks

IAMs typically describe energy transitions in terms of technology deployment, capacity expansion, and cumulative investment requirements, rather than sectoral production or interindustry flows. In line with a well-established literature, the analysis adopts a soft-linking strategy, whereby IAM outputs are translated into reallocations of final demand within the IO system. This approach is widely used in the literature to combine the technological detail of IAMs with the economy-wide structural resolution of IO models, while avoiding the strong equilibrium assumptions required by computable general equilibrium frameworks (Merciai and Heijungs, 2014; Wiebe et al., 2019; Pollitt et al., 2021; Vrontisi et al., 2022).

Specifically, BLUES is used to derive time profiles for cumulative transition effort associated with broad mitigation categories, including renewable electricity generation, electricity grids and storage, transport electrification, industrial energy efficiency, and CDR options such as bioenergy with carbon capture and storage (BECCS). These trajectories capture the timing and intensity of structural change required to meet Brazil's NDC targets.

Translation of IAM outputs into the IO framework follows a two-step procedure. First, IAM mitigation categories are mapped onto existing sectors in the Brazilian input–output table using fixed allocation coefficients that reflect the composition of capital goods, infrastructure, and services associated with transition-related activities. These mappings primarily involve construction, machinery and equipment manufacturing, metals, utilities, transport equipment, and technical and professional services. Fossil-related sectors — particularly crude oil and natural gas extraction and petroleum refining — experience a relative contraction compared to

the baseline trajectory. Emerging activities such as hydrogen production and CDR are not introduced as new sectors; instead, they are represented through increased demand for existing industrial, energy, and service sectors embedded in the observed production structure.

Second, scenario-specific final demand shocks are constructed as reallocations of demand across this subset of transition-relevant sectors. For each year t, final demand is defined as:

$$\mathbf{y}_t^{(s)} = \mathbf{y}_t^{\text{baseline}} + \Delta\mathbf{y}_t^{(s)}$$

where $\mathbf{y}_t^{\text{baseline}}$ represents the baseline final demand trajectory for all sectors, and $\Delta\mathbf{y}_t^{(s)}$ is a sparse vector capturing the scenario-specific reallocation of demand across a subset of transition-relevant sectors. For non-targeted sectors, $\Delta y_{i,t}^{(s)} = 0$, implying that they follow the baseline path. As such, this formulation ensures that differences between scenarios arise from the timing and sectoral composition of transition-related demand.

2.4. Scenario design and assessed outcomes

The analysis considers two core scenarios over the period 2025–2050: a Baseline scenario and an NDC-Transition scenario. The Baseline scenario represents a continuation of current production and demand patterns in the absence of additional climate policy beyond those already established. To provide a consistent macroeconomic envelope for the IO simulations, aggregate final demand is assumed to grow at a constant annual rate of 1.5%. This conservative long-term growth assumption is consistent with medium-term macroeconomic outlooks for Brazil and is commonly adopted in structural scenario analyses to avoid overstating transition effects (IMF, 2023; World Bank, 2023). Growth is applied proportionally across all sectors, keeping the sectoral composition of demand constant over time.

The NDC-Transition scenario reflects a reallocation of final demand consistent with Brazil's mitigation commitments, as informed by BLUES. In this scenario, the same aggregate growth envelope is maintained, but the composition of final demand is altered through a time-dependent vector of transition shocks, $\Delta \mathrm{y}_t$, applied to a subset of transition-relevant sectors. In practical terms, the transition increases demand for infrastructure and capital-intensive activities (e.g., construction, machinery and equipment, utilities, metals, and technical and professional services) while reducing demand for fossil-related sectors (notably crude oil and gas extraction and petroleum refining) relative to the baseline path. Carbon dioxide removal options are represented indirectly through higher demand for existing sectors supplying bioenergy, electricity, industrial equipment, engineering services, and supporting infrastructure, particularly as the economy approaches net-zero by mid-century.

Scenario outcomes are evaluated using a set of macro-structural and socio-economic indicators derived from the demand-driven IO model. At the macro-structural level, we assess changes in sectoral gross output and total economy-wide

using the Leontief inverse to capture direct and indirect effects across the production network. At the socio-economic level, we examine employment outcomes and labour income. This design enables a structural and socio-economic assessment of an NDC-aligned energy transition, emphasizing how production and labour market outcomes evolve as the economy absorbs large-scale structural reallocation.

3. Results

3.1 Baseline Production Structure

Figure 1 summarizes production, employment, and income multipliers, together with forward linkage extraction losses, revealing the structural channels through which demand shocks propagate in the Brazilian economy under baseline conditions.

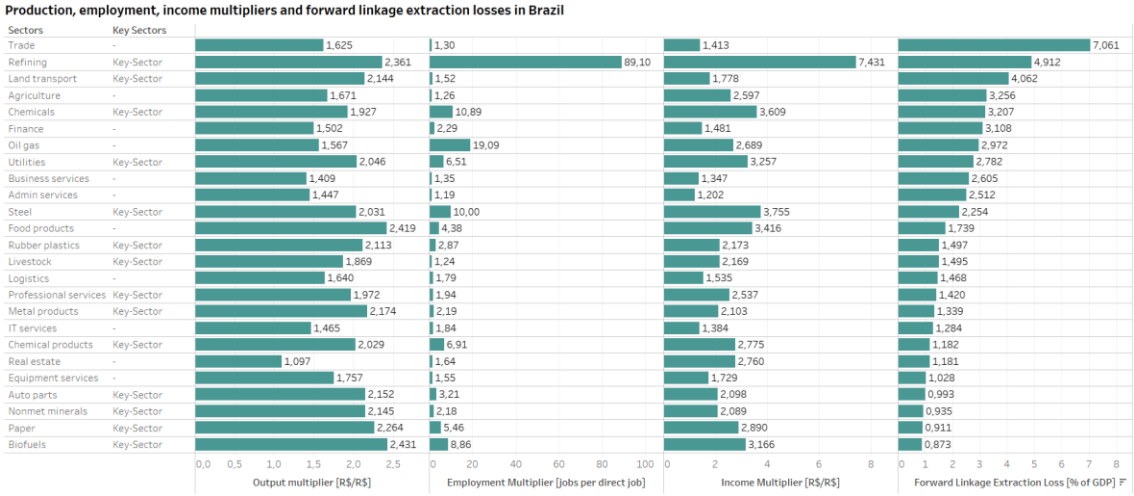

Figure 1. Structural multipliers and forward linkage extraction losses in the Brazilian economy under baseline conditions (2021).

Production multipliers (MP) range from approximately 1.1 to 2.4, indicating substantial heterogeneity in upstream production linkages. The highest output multipliers are observed in biofuels (2.43), food products (2.42), petroleum refining (2.36), paper (2.26), steel (2.03), chemical products (2.03), and energy utilities (2.05). These sectors display dense intermediate input structures, meaning that a one-real increase in final demand generates more than two reais of total gross output once indirect effects are accounted for. By contrast, service-oriented activities such as real estate (1.10), business services (1.41), administrative services (1.45), and IT services (1.47) exhibit lower production multipliers, reflecting shorter supply chains and a higher reliance on primary inputs rather than intermediate production.

Employment multipliers per direct job (MEI) exhibit even stronger dispersion, ranging from values close to 1.2 up to almost 90 jobs per direct job. Petroleum refining stands out as an extreme case, with an employment multiplier of approximately 89, indicating that each direct job in refining sustains a very large

number of jobs elsewhere in the economy. Similarly, high employment multipliers are observed in oil and gas extraction (19), chemicals (10.9), steel (10.0), biofuels (8.9), utilities (6.5), and paper production (5.5). In contrast, labor-intensive service sectors such as agriculture (1.26), livestock (1.24), administrative services (1.19), and business services (1.35) display low employment multipliers, indicating that employment effects are concentrated primarily in the sector itself, with limited indirect propagation.

Income multipliers (MRI), measured as total labor income generated per unit of direct labor income, range from approximately 1.2 to 7.4. Again, petroleum refining exhibits the highest value (7.43), followed by steel (3.76), food products (3.42), biofuels (3.17), utilities (3.26), and chemicals (3.61). These values indicate that although direct wage payments in these sectors are relatively small, each unit of direct labor income triggers substantial wage generation along upstream and induced channels. Service sectors, including IT services (1.38), equipment services (1.73), and real estate (2.76), show more limited income propagation, consistent with weaker intersectoral wage spillovers.

Forward linkage extraction losses (FL%) provide a measure of systemic importance by capturing the contraction in total gross output that results from the hypothetical removal of a sector as a supplier. Trade exhibits the largest extraction loss, at approximately 7.1% of total gross output, underscoring its role as a universal intermediary across production chains. Petroleum refining (4.9%), land transport (4.1%), agriculture (3.3%), chemicals (3.2%), and finance (3.1%) also exhibit significant forward losses, indicating that disruptions in these sectors would have substantial economy-wide effects. By contrast, manufacturing niches such as biofuels (0.87%), paper (0.91%), and non-metallic minerals (0.94%) exhibit lower forward extraction losses, despite high production multipliers, reflecting a stronger upstream than downstream systemic role.

These results reveal a production network in which capital-intensive and energy-related sectors occupy central positions, despite their relatively low direct employment and wage intensity. Petroleum refining and oil and gas extraction, in particular, combine moderate production multipliers, high employment and income multipliers, and large forward linkage losses, highlighting their dual role as macro-structural amplifiers.

3.2 NDC-Transition vs Baseline: Structural reallocation effects

Figure 2 reports the deviation of total gross output under the NDC-Transition scenario relative to the baseline. At the aggregate level, the transition is associated with a positive and increasing output differential over time. Total gross output is estimated to be approximately 0.3% higher than baseline in 2030, 0.8% higher in 2040, and 1.2% higher in 2050.

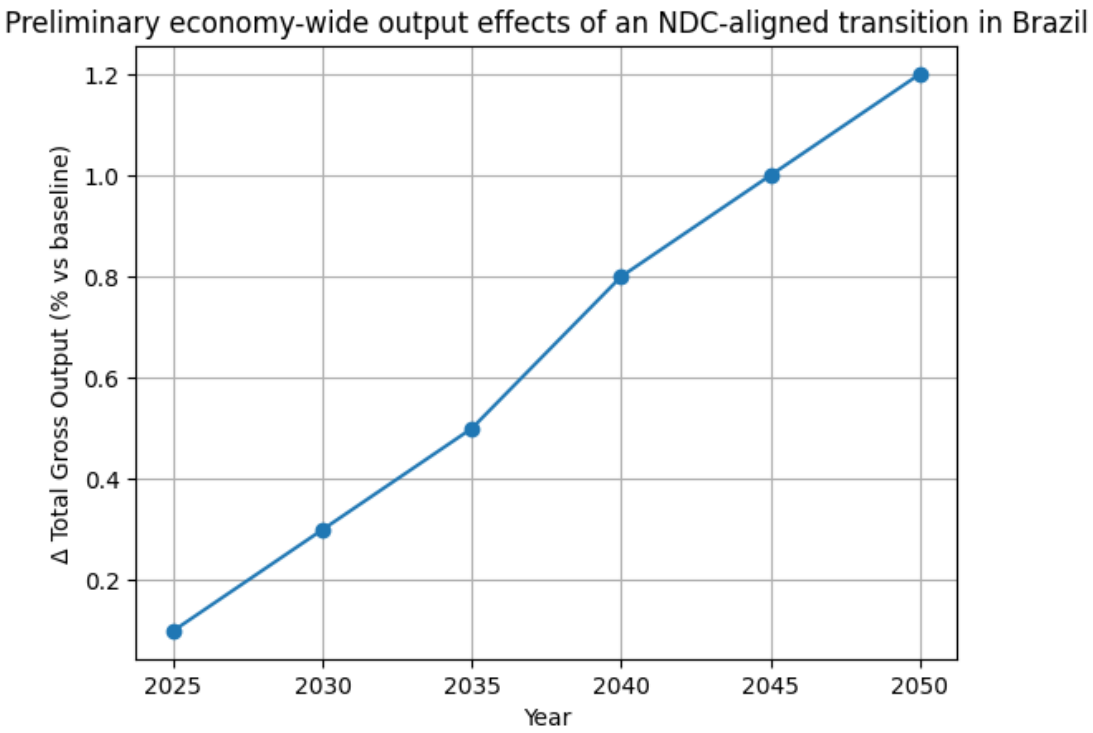

Figure 2. Preliminary economy-wide deviation of total gross output under an NDC-aligned transition relative to the baseline scenario in Brazil (2025–2050).

These preliminary results indicate that, given the current structure of intersectoral linkages in the Brazilian economy, reallocating demand toward transition-relevant activities generates net expansionary effects. The mechanism underlying this outcome is structural rather than macro-demand driven: sectors receiving positive demand shocks under the NDC scenario are characterized by relatively high backward linkages and strong indirect effects, which amplify the initial impulse through domestic supply chains.

Sectoral results for 2050 highlight a clear reorientation of economic activity. Positive deviations relative to baseline are concentrated in construction (+4.5%), machinery and equipment manufacturing (+3.8%), utilities (+3.2%), basic metals (+2.9%), and professional and technical services (+2.6%). These sectors play a central role in supplying capital goods, infrastructure, and enabling services required for the energy transition.

Conversely, oil and gas extraction (–3.6%) and petroleum refining (–4.1%) experience the largest negative deviations. These contractions reflect the declining role of fossil-based activities in the NDC-consistent demand composition and their limited capacity to benefit from transition-related investment spillovers within the existing production structure.

Preliminary estimates of socio-economic impacts indicate that the structural reallocation associated with the NDC-Transition scenario is also reflected in labor market outcomes. Relative to the baseline, total employment is estimated to increase by approximately 180 thousand jobs in 2030 and 420 thousand jobs in 2050.

Employment gains are primarily concentrated in sectors with higher labor intensity and strong upstream integration, including construction, manufacturing of machinery and equipment, transport and logistics, and business and professional services. In contrast, employment losses in fossil-related sectors are limited in absolute terms, reflecting their capital-intensive production structures and relatively low direct employment per unit of output.

Aggregate labor income follows a similar pattern. Total wage income under the NDC-Transition scenario is estimated to be 0.6% higher than baseline in 2030 and 1.1% higher in 2050. While sectors such as oil refining and mining exhibit low direct wage intensity, their income multipliers imply that changes in these activities continue to generate indirect income effects across the economy. Nevertheless, net income gains are dominated by expansion in labor-intensive service and construction-related activities.

4. Concluding remarks

This study shows that the economic implications of Brazil's NDC are shaped by the structure of intersectoral linkages in the production network. Under current technological and institutional conditions, sectors that expand in an NDC-aligned transition, such as construction, machinery, utilities, and professional services, are strongly embedded in upstream supply chains. As a result, the reorientation of economic activity required to meet climate targets propagates broadly through the economy, yielding positive aggregate output, employment, and income effects relative to a baseline trajectory.

At the same time, the results highlight a key structural challenge: fossil-related sectors remain systemically important despite their declining role in the transition pathway. Oil and gas extraction and petroleum refining combine low direct labor intensity with high income multipliers and significant forward linkage losses, implying that adjustments in these sectors generate wide indirect effects. This underscores the need to complement climate policy with industrial and labor-market strategies that account for structural dependencies in fossil-related sectors, so that NDC implementation remains economically and socially manageable.

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